

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 6

E-UTRAN — LTE Radio Access Network

E-UTRAN — شبکه دسترسی رادیویی LTE



Lecture Notes

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6.1 E-UTRAN Architecture

Why No RNC? The Flat Architecture Decision

In UMTS (3G), the Radio Network Controller (RNC) sat between the NodeBs and the Core Network, centralizing radio resource management, handover decisions, and protocol termination. LTE **eliminated the RNC entirely**, creating a flat, single-node radio access network.

Reasons for removing the RNC:

Concern	3G (with RNC)	LTE (flat, no RNC)
Latency	Extra hop through RNC adds ~5–10 ms	Direct eNB↔EPC path, sub-5 ms user-plane latency
Single point of failure	RNC failure affects all connected NodeBs	Each eNB is independent
Scalability	RNC capacity limits number of NodeBs	No bottleneck; eNBs scale independently
Complexity	Three-node RAN (NodeB→RNC→CN)	Two-node RAN (eNB→EPC)
Cost	Expensive, proprietary RNC hardware	Distributed intelligence reduces CAPEX
Handover speed	RNC-mediated = slower	Direct eNB-to-eNB via X2 = faster

Key architectural principle: All radio-related intelligence is pushed into the **eNodeB (eNB)**, which communicates directly with the Evolved Packet Core (EPC) via the S1 interface and with neighboring eNBs via the X2 interface.

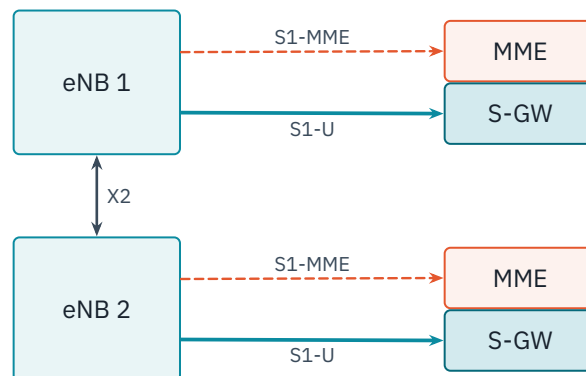


Figure 6.1 Each eNB connects to the MME over S1-MME and to the S-GW over S1-U

6.2 eNodeB Functions

The eNodeB is the **sole node** in E-UTRAN. It performs all functions previously split between NodeB and RNC:

Radio Resource Management (RRM)

- **Radio Bearer Control:** Establishment, maintenance, and release of radio bearers
- **Radio Admission Control (RAC):** Accept/reject new radio bearer requests based on interference and load
- **Connection Mobility Control:** Handover decisions based on UE measurements
- **Dynamic Resource Allocation (Scheduling):** Assigns resource blocks to UEs every TTI (1 ms)

Scheduling

- Operates every **1 ms TTI** (Transmission Time Interval)
- Frequency-domain and time-domain scheduling
- Algorithms: Round Robin, Proportional Fair, Max Throughput
- Uses **CQI** (Channel Quality Indicator) reports from UEs
- Controls both DL and UL resource allocation

HARQ (Hybrid Automatic Repeat Request)

- Stop-and-wait HARQ with **8 parallel processes**
- Synchronous in UL, asynchronous in DL
- Soft combining: Chase Combining or Incremental Redundancy
- Round-trip time: 8 ms (4 ms each way)

Channel Encoding

- **Turbo coding** (rate 1/3) for data channels
- **Convolutional coding** for control channels
- Rate matching and HARQ buffer management
- CRC attachment for error detection

Other Functions

- IP header compression (ROHC)
- Cipherring and integrity protection
- Paging message distribution
- Broadcast information (SIBs, MIBs)
- Measurement configuration and reporting

6.3 LTE Protocol Stack

Protocol Stack Diagram

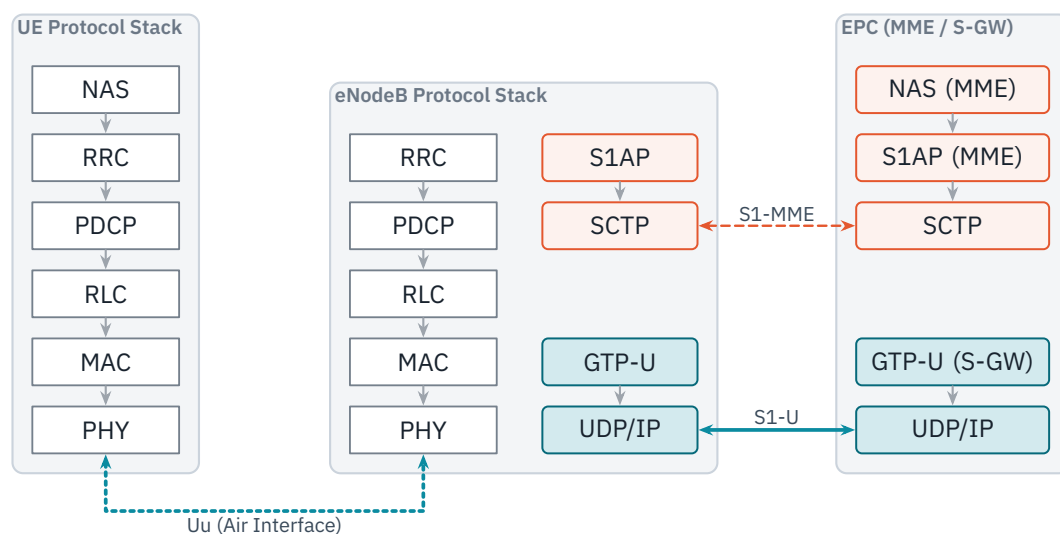


Figure 6.2 LTE protocol stacks of the UE, the eNodeB and the EPC

Control Plane Stack

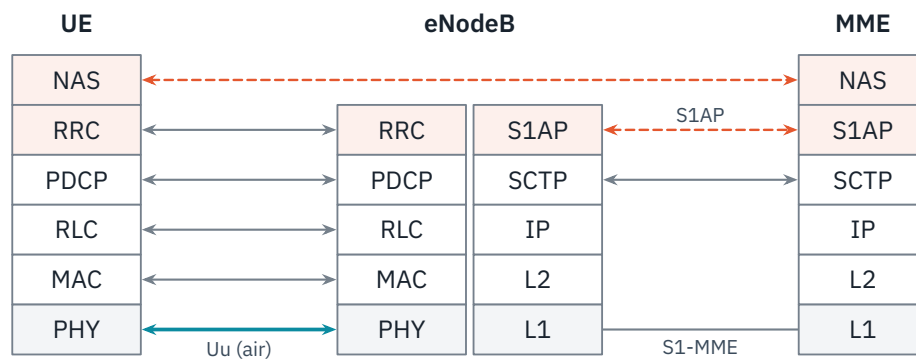


Figure 6.3 LTE control-plane protocol stack

User Plane Stack

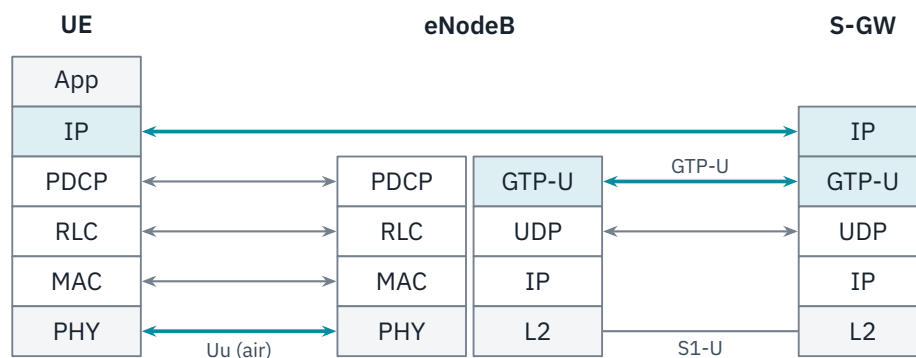


Figure 6.4 LTE user-plane protocol stack

6.4 Protocol Layers in Detail

6.4.1 Physical Layer (PHY)

■ Downlink: OFDMA (Orthogonal Frequency Division Multiple Access)

- Divides bandwidth into many narrow **subcarriers** (15 kHz spacing)
- Each subcarrier is orthogonal → no inter-carrier interference
- Robust against multipath fading (cyclic prefix absorbs delay spread)
- Enables flexible frequency-domain scheduling

Resource Grid:

- **Resource Element (RE):** 1 subcarrier × 1 OFDM symbol = smallest allocatable unit
- **Resource Block (RB):** 12 subcarriers × 7 OFDM symbols (1 slot = 0.5 ms) = **84 REs**
- **Resource Block Pair:** 2 RBs = 1 subframe (1 ms TTI)

Bandwidth	Number of RBs	Subcarriers
1.4 MHz	6	72
3 MHz	15	180
5 MHz	25	300
10 MHz	50	600
15 MHz	75	900
20 MHz	100	1200

Frame Structure (FDD):

- 1 radio frame = 10 ms = 10 subframes
- 1 subframe = 1 ms = 2 slots
- 1 slot = 0.5 ms = 7 OFDM symbols (normal CP)

■ Uplink: SC-FDMA (Single Carrier FDMA)

- Also called DFT-spread OFDM (DFT-s-OFDM)
- Lower **PAPR** (Peak-to-Average Power Ratio) than OFDMA → better for battery-powered UEs
- Single-carrier property achieved by DFT precoding before IFFT
- Same resource grid structure as DL (12 subcarriers per RB)
- **Constraint:** UL allocation must be contiguous in frequency

■ Key PHY Procedures

- Cell search (PSS/SSS synchronization)
- Channel estimation (reference signals: CRS, DM-RS, SRS)
- MIMO: up to 4×4 in DL, 1×2 or 2×2 in UL (Rel-8/10)
- Link adaptation via CQI/PMI/RI feedback
- Power control (open-loop and closed-loop)

6.4.2 MAC Layer (Medium Access Control)

The MAC layer sits between RLC and PHY, responsible for **real-time scheduling** and data multiplexing.

■ Key Functions

1. Scheduling (DL and UL)

- Operates every 1 ms TTI
- Assigns RBs based on CQI, buffer status, QoS priority, fairness
- DL: eNB decides; UL: eNB grants resources to UEs

2. Multiplexing/Demultiplexing

- Multiplexes data from multiple logical channels into one transport block
- Uses Logical Channel Prioritization (LCP) in UL

3. HARQ Entity

- 8 parallel stop-and-wait processes
- DL: Asynchronous adaptive HARQ
- UL: Synchronous HARQ (fixed timing)
- Generates ACK/NACK; soft combining at receiver

4. Random Access Procedure

- Contention-based (4-step) or contention-free (2-step)
- Used for initial access, handover, UL synchronization recovery

5. BSR (Buffer Status Report)

- UE informs eNB about pending UL data
- Triggers UL scheduling grants

6. PHR (Power Headroom Report)

- UE reports remaining power margin
- Helps eNB with UL scheduling decisions

■ Channel Mapping at MAC

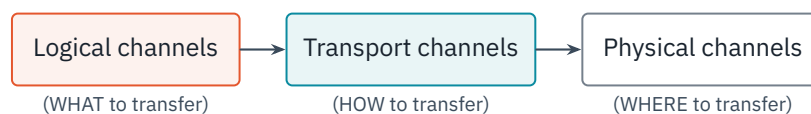


Figure 6.5 Three levels of LTE channels

6.4.3 RLC Layer (Radio Link Control)

RLC provides segmentation, reassembly, and (optionally) ARQ.

■ Three Modes

Mode	Full Name	ARQ	Segmentation	Use Case
TM	Transparent Mode	No	No	BCCH, PCCH, CCCH (small, fixed-size messages)
UM	Unacknowledged Mode	No	Yes	VoIP, real-time video (delay-sensitive)
AM	Acknowledged Mode	Yes	Yes	TCP traffic, signaling (reliability needed)

■ Key Functions

- **Segmentation and Reassembly:** Splits PDCP PDUs into RLC-sized segments fitting transport block size
- **Concatenation:** Packs multiple small SDUs into one PDU

- **ARQ (AM only):** Retransmits lost RLC PDUs based on STATUS reports
 - Polling mechanism: RLC sender requests STATUS from receiver
 - Operates independently from HARQ (RLC ARQ catches what HARQ misses)
- **Reordering (UM/AM):** Delivers SDUs in order to upper layers
- **Duplicate Detection (AM):** Discards duplicate PDUs

■ RLC vs. HARQ

- HARQ: Fast (8 ms RTT), limited retransmissions (typically 4)
- RLC ARQ: Slower, but catches residual errors after HARQ exhaustion
- Combined: achieves target BLER of 10^{-6} for AM bearers

6.4.4 PDCP Layer (Packet Data Convergence Protocol)

PDCP sits above RLC, providing security and efficiency functions.

■ Key Functions

1. Header Compression (User Plane only)

- Uses **ROHC** (Robust Header Compression, RFC 3095/4995)
- Compresses IP/UDP/RTP headers from ~40–60 bytes → 1–4 bytes
- Critical for VoIP: 40-byte header on 32-byte payload → ~56% overhead without ROHC
- Multiple ROHC profiles for different protocol stacks

2. Ciphering (User Plane + Control Plane)

- Encrypts all user data and signaling
- Algorithms: EEA0 (null), 128-EEA1 (SNOW 3G), 128-EEA2 (AES-CTR), 128-EEA3 (ZUC)
- Uses COUNT (sequence number + HFN) + BEARER + DIRECTION as input

3. Integrity Protection (Control Plane only)

- Protects RRC signaling from tampering
- Algorithms: EIA1 (SNOW 3G), EIA2 (AES-CMAC), EIA3 (ZUC)
- Generates 32-bit MAC-I appended to each RRC message
- **Not applied to user plane in LTE** (added in 5G NR for user plane)

4. Sequence Numbering and Reordering

- Assigns PDCP SN to each SDU
- Reorders out-of-sequence PDUs (important during handover)
- Detects duplicates

5. Handover Support

- In-sequence delivery and duplicate elimination during handover
- Status reporting: tells source eNB which PDUs were successfully delivered
- Enables lossless handover for AM bearers

■ PDCP PDU Format

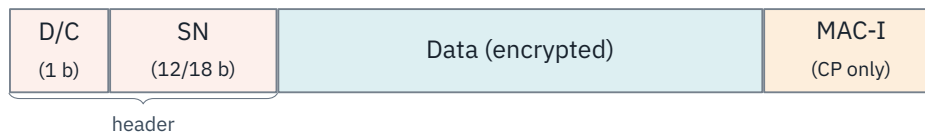


Figure 6.6 PDCP PDU format

6.4.5 RRC Layer (Radio Resource Control)

RRC is the **control plane protocol** between UE and eNodeB, managing the radio connection.

■ Key Functions

1. Connection Management

- RRC Connection Setup / Reconfiguration / Release
- Security activation (ciphering + integrity)
- SRB (Signaling Radio Bearer) establishment

2. Measurement Configuration and Reporting

- Configures UE to measure serving and neighbor cells
- Events: A1–A6 (intra/inter-frequency), B1–B2 (inter-RAT)
- Reporting: periodic or event-triggered
- Quantities: RSRP, RSRQ, SINR

3. Handover Control

- eNB makes handover decision based on measurement reports
- Sends **RRCConnectionReconfiguration** with mobility control info
- UE performs handover, sends **RRCConnectionReconfigurationComplete**

4. System Information Broadcasting

- **MIB**: Transmitted on PBCH every 40 ms (bandwidth, SFN, PHICH config)
- **SIB1**: Cell access info, scheduling of other SIBs (80 ms period)
- **SIB2**: Common channel config (RACH, paging, UL power control)
- **SIB3–SIB8**: Cell reselection, inter-frequency/inter-RAT info
- **SIB9–SIB19**: Additional info (home eNB, MBMS, etc.)

5. Paging

- Notifies UEs in IDLE of incoming calls/data
- DRX-based: UE wakes at specific paging occasions

6. NAS Message Transfer

- Transparently carries NAS messages between UE and MME

■ Signaling Radio Bearers (SRBs)

SRB	Purpose	RLC Mode
SRB0	RRC messages on CCCH (before security)	TM
SRB1	RRC messages + piggybacked NAS (after setup)	AM
SRB2	NAS messages (lower priority than SRB1)	AM

6.5 LTE Channel Structure

■ Channel Types Overview

LTE defines three categories of channels, each at a different layer of abstraction:

Category	Layer	Defined By	Question Answered
Logical Channels	MAC (upper)	What type of data	What is being transferred?
Transport Channels	MAC (lower)	How data is transferred	How is it characterized?
Physical Channels	PHY	Where in the resource grid	Where are the bits placed?

■ Logical Channels

■ Control Channels

Channel	Name	Direction	Purpose
BCCH	Broadcast Control Channel	DL	System information (MIB, SIBs)
PCCH	Paging Control Channel	DL	Paging messages for IDLE UEs
CCCH	Common Control Channel	DL/UL	Messages before RRC connection (Msg3/Msg4)
DCCH	Dedicated Control Channel	DL/UL	Dedicated signaling (RRC after connection)
MCCH	Multicast Control Channel	DL	MBMS control information

■ Traffic Channels

Channel	Name	Direction	Purpose
DTCH	Dedicated Traffic Channel	DL/UL	User data (IP packets)
MTCH	Multicast Traffic Channel	DL	MBMS user data

■ Transport Channels

Channel	Name	Direction	Purpose
BCH	Broadcast Channel	DL	Carries MIB; fixed format, 40 ms period
PCH	Paging Channel	DL	Carries paging; supports DRX
DL-SCH	Downlink Shared Channel	DL	Main DL data; supports HARQ, dynamic scheduling
UL-SCH	Uplink Shared Channel	UL	Main UL data; supports HARQ, dynamic scheduling
RACH	Random Access Channel	UL	Random access preambles
MCH	Multicast Channel	DL	MBMS data; fixed scheduling

■ Physical Channels

■ Downlink Physical Channels

Channel	Name	Purpose
PDSCH	Physical DL Shared Channel	User data + system info (SIBs, paging)
PDCCH	Physical DL Control Channel	DCI (scheduling grants, power control)
PBCH	Physical Broadcast Channel	MIB transmission
PMCH	Physical Multicast Channel	MBMS data
PCFICH	Physical CFI Channel	Indicates # of OFDM symbols for PDCCH
PHICH	Physical HARQ Indicator Channel	UL HARQ ACK/NACK

■ Uplink Physical Channels

Channel	Name	Purpose
PUSCH	Physical UL Shared Channel	UL user data + control info
PUCCH	Physical UL Control Channel	CQI, HARQ ACK/NACK, SR
PRACH	Physical Random Access Channel	Random access preambles

■ Channel Mapping

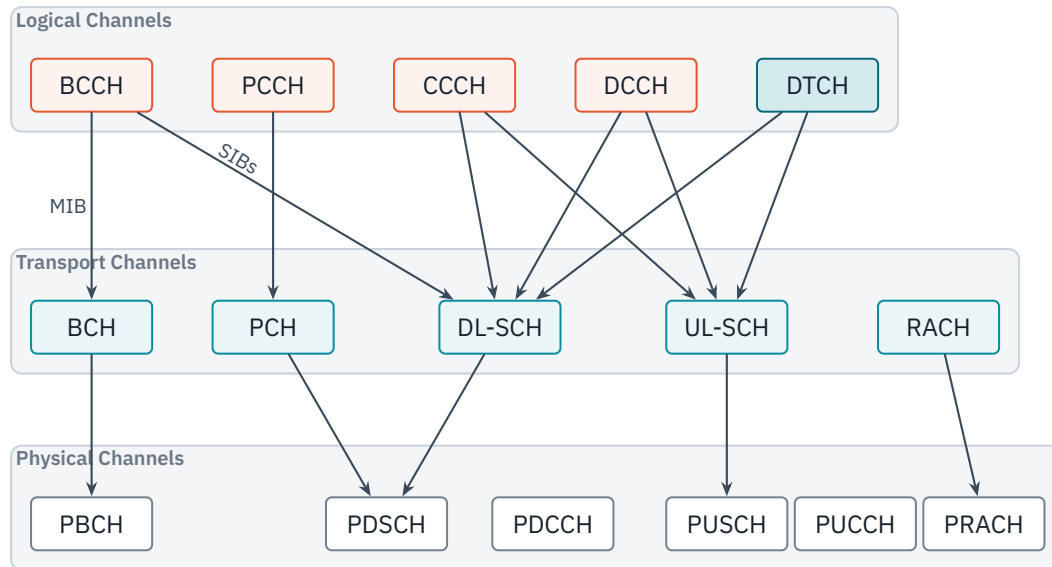


Figure 6.7 Mapping of logical, transport and physical channels

Complete Channel Mapping Table

Logical Channel	Transport Channel	Physical Channel	Direction
BCCH (MIB)	BCH	PBCH	DL
BCCH (SIBs)	DL-SCH	PDSCH	DL
PCCH	PCH	PDSCH	DL
CCCH	DL-SCH	PDSCH	DL
CCCH	UL-SCH	PUSCH	UL
DCCH	DL-SCH	PDSCH	DL
DCCH	UL-SCH	PUSCH	UL
DTCH	DL-SCH	PDSCH	DL
DTCH	UL-SCH	PUSCH	UL
— (HARQ feedback)	—	PHICH	DL
— (DCI/grants)	—	PDCCH	DL
— (CFI)	—	PCFICH	DL
— (CQI/SR/ACK)	—	PUCCH	UL
— (RA preamble)	RACH	PRACH	UL

6.6 S1 Interface

The S1 interface connects the eNodeB to the EPC. It is split into:

- **S1-MME (S1-C):** Control plane → eNB ↔ MME
- **S1-U:** User plane → eNB ↔ S-GW

S1-MME Protocol Stack

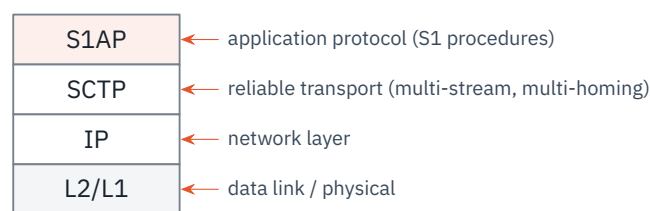


Figure 6.8 S1-MME protocol stack

Why SCTP (not TCP)?

- Multi-streaming: avoids head-of-line blocking
- Multi-homing: failover between IP addresses
- Message-oriented (preserves boundaries)

S1AP Procedures

UE-Associated Procedures (per-UE context)

Procedure	Direction	Purpose
Initial UE Message	eNB → MME	Carries first NAS message (Attach, Service Request)
Downlink NAS Transport	MME → eNB	Delivers NAS message to UE
Uplink NAS Transport	eNB → MME	Forwards NAS message from UE
Initial Context Setup	MME → eNB	Establishes security context + default bearer
UE Context Release	MME ↔ eNB	Releases UE context (Request/Command/Complete)
Handover Preparation	Source eNB → MME	S1-based handover when X2 unavailable
Handover Resource Allocation	MME → Target eNB	Allocates resources at target
Path Switch Request	Target eNB → MME	Updates S-GW path after X2 handover
E-RAB Setup/Modify/Release	MME → eNB	Bearer management

Non-UE-Associated Procedures (global)

Procedure	Purpose
S1 Setup	eNB registers with MME (TAC, PLMN, capacity)
Reset	Error recovery (partial or full)
Paging	MME sends paging to eNBs in tracking area
eNB/MME Configuration Update	Capacity/configuration changes
Overload Start/Stop	MME signals congestion
Error Indication	Report protocol errors

S1-U Protocol Stack

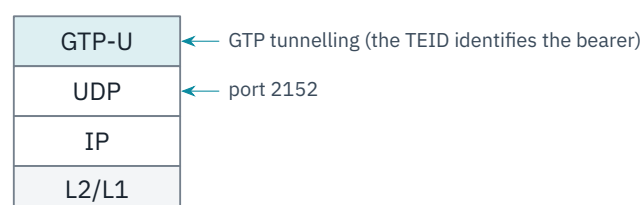


Figure 6.9 S1-U protocol stack

- Each E-RAB has a **GTP tunnel** identified by TEID (Tunnel Endpoint ID)
- One-to-one mapping: EPS bearer ↔ S1 GTP tunnel ↔ radio bearer

6.7 X2 Interface

The X2 interface connects **neighboring eNodeBs** directly for:

- Fast handover (no MME involvement for intra-LTE mobility)
- Inter-cell load management
- Inter-cell interference coordination (ICIC)

■ X2 Protocol Stack

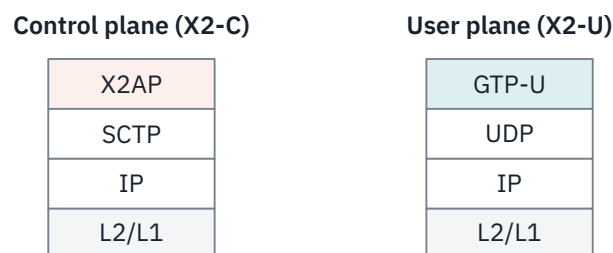


Figure 6.10 X2 protocol stacks

■ X2AP Procedures

■ Mobility Management (Handover)

X2 Handover Sequence:

1. **Source eNB** sends **Handover Request** → Target eNB (includes UE context, bearer info)
2. **Target eNB** performs admission control → sends **Handover Request Acknowledge**
3. **Source eNB** sends **RRConnectionReconfiguration** to UE (with target cell info)
4. **UE** synchronizes to target cell, sends **RRConnectionReconfigurationComplete**
5. **Target eNB** sends **Path Switch Request** to MME (updates core network path)
6. **Source eNB** receives **UE Context Release** → releases resources

During handover: Data forwarded source→target via X2-U GTP tunnel (avoids data loss).

■ Load Management

Procedure	Purpose
Resource Status Reporting	Request/report load information (PRB usage, composite load)
Load Indication	Inform neighbor about UL interference (HII, OI)
Mobility Change Request	Negotiate handover parameters (CIO adjustment)

■ Inter-Cell Interference Coordination (ICIC)

- **Load Information message** carries:
 - **HII (High Interference Indication)**: UL RBs where cell plans to schedule cell-edge UEs
 - **OI (Overload Indicator)**: UL RBs experiencing high interference
 - **RNTP (Relative Narrowband TX Power)**: DL RBs where cell will exceed power threshold

■ X2 Setup and Configuration

- X2 link established via **X2 Setup procedure** (or via MME using S1 ENB Configuration Transfer)
- Automatic Neighbor Relation (ANR): eNB discovers neighbors and establishes X2
- Multiple eNBs can connect forming a mesh topology

6.8 RRC States and Transitions

■ RRC State Machine

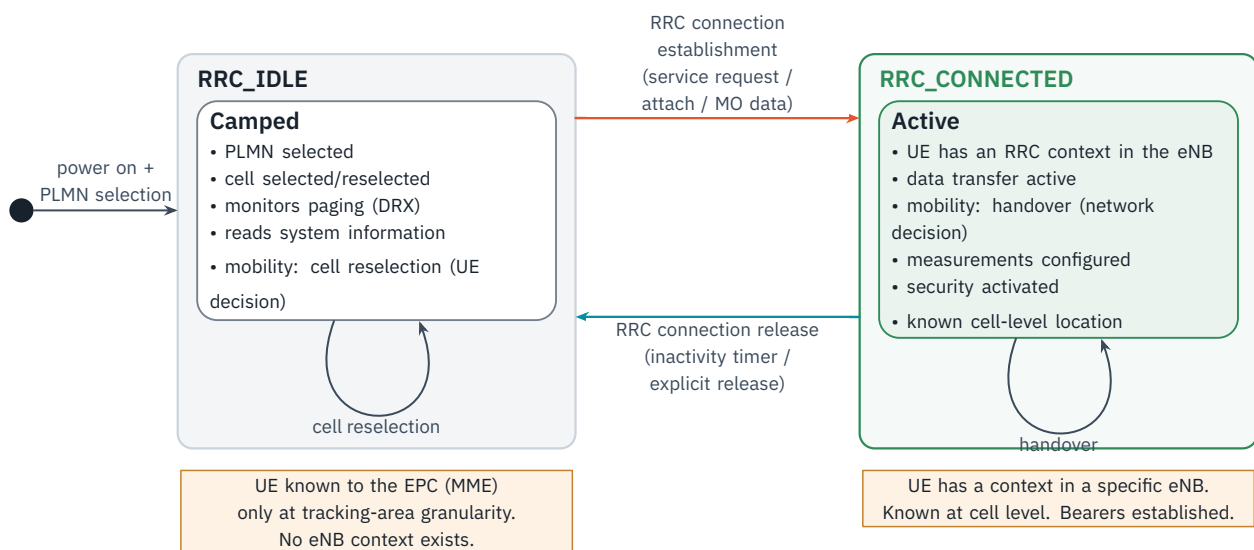


Figure 6.11 LTE RRC state machine

■ RRC_IDLE State

Characteristic	Description
Context	No eNB context; registered with MME only
Location	Known at Tracking Area level
Mobility	Cell reselection (UE-controlled, based on SIBs)
Data	No data transfer possible
Paging	UE monitors paging channel with DRX cycle
System Info	UE acquires MIB and relevant SIBs
Power	Low power consumption (long DRX sleep)

■ RRC_CONNECTED State

Characteristic	Description
Context	Full UE context in serving eNB
Location	Known at cell level
Mobility	Handover (network-controlled, measurement-based)
Data	Active data transfer (DL + UL)
Measurements	Configured by eNB (events A1–A6, B1–B2)
DRX	Connected-mode DRX possible (shorter cycles)
Security	AS ciphering + integrity active

■ State Transitions

■ IDLE → CONNECTED (Connection Establishment)

1. UE sends **RRC Connection Request** (on CCCH via SRB0)
2. eNB responds with **RRC Connection Setup** (configures SRB1)
3. UE sends **RRC Connection Setup Complete** (carries initial NAS message)
4. eNB forwards NAS to MME via S1AP **Initial UE Message**
5. MME initiates **Initial Context Setup** (security + bearer establishment)

■ CONNECTED → IDLE (Connection Release)

- MME triggers via S1AP **UE Context Release Command**
- eNB sends **RRC Connection Release** to UE
- UE deletes AS context, returns to idle
- Triggers: inactivity timer, explicit NAS detach, radio link failure recovery failure

5G Extension: RRC_INACTIVE State

In 5G NR (and late LTE Rel-15+), a third state is introduced:

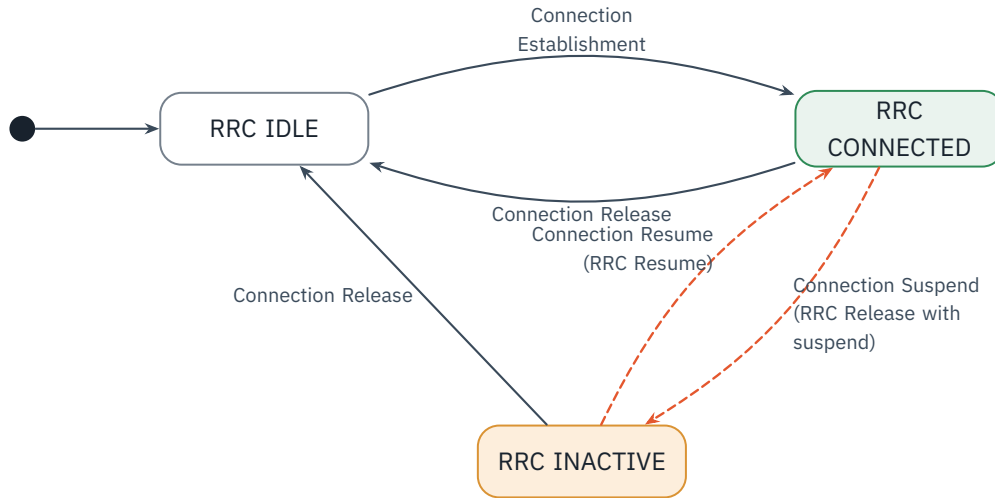


Figure 6.12 5G NR RRC states with RRC_INACTIVE

State	UE Context	Location Granularity	Mobility	CN Connection
IDLE	None in RAN	Tracking Area	Cell reselection	None
INACTIVE	Stored in gNB (suspended)	RAN Notification Area	Cell reselection + RAN paging	Maintained with CN
CONNECTED	Active in gNB	Cell level	Handover	Active

Benefits of INACTIVE state:

- Faster resumption than IDLE→CONNECTED (skip NAS procedures)
- Power saving (no connected-mode signaling)
- Reduced signaling load on core network
- AS context preserved → faster resume with single RRC Resume message
- Ideal for IoT devices with intermittent small data transmissions

Summary

Component	Key Takeaway
Architecture	Flat (no RNC); eNB handles all RAN functions
eNodeB	Scheduling, RRM, HARQ, handover decisions, encoding
PHY	OFDMA (DL) / SC-FDMA (UL); resource blocks as basic unit
MAC	1 ms scheduling, HARQ, multiplexing, random access
RLC	TM/UM/AM; segmentation + ARQ for reliability
PDCP	ROHC compression, ciphering, integrity (CP), reordering
RRC	Connection/mobility management, measurement, system info
Channels	Logical→Transport→Physical mapping; shared channels dominate
S1	eNB↔EPC; S1AP (SCTP) for control, GTP-U for user plane
X2	eNB↔eNB; fast handover, load mgmt, interference coordination
RRC States	IDLE (low power) ↔ CONNECTED (active); INACTIVE in 5G

Module 6 — E-UTRAN (LTE Radio Access Network) — Advanced Communication Networks